

Final Report

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1 Introduction

In this report, we have all of the possible colorings for P_4 , i.e. path graphs with 4 vertices, and we classify them according to whether their vanishing ideals are toric (after the change of variables).

For the Macaulay2 and SageMath Codes, please see the Github repository: github.com/tqueer/toric. Please also see [1] for reference.

2 Colored Path Graphs with 4 Vertices

2.1 All of the 44 Colorings.

	Edge Type A	Dim	Edge Type B	Dim	Edge Type C	Dim	Edge Type D	Dim
1		3		1		2		37
2		14		14		4		37
3		15		14		4		37
4		14		1		2		37
5		14		2		3		25
6		14		53		2		43
7		2		13		2		43
8		21		2		8		14
9		2		2		3		13
10		3		2		4		14
11		8		8		13		3
12		25		42		12		64

Toric without the change of variables

Toric under the change of variables

Dim = Dimension of the Lie Albra

Figure 1: Classification table for T_4

2.2 The 12 Toric Colorings of P_4 Graphs.

1. PA5 (Dimension of Lie Algebra: 14.)

The list of generators:

$$\begin{aligned}
& s_{1,1} - s_{2,2} + s_{1,3}, \\
& s_{3,3}s_{2,4} - s_{2,3}s_{3,4}, \\
& s_{1,4}^2 - s_{2,4}^2 + s_{1,4}s_{3,4}, \\
& s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\
& s_{2,3}s_{1,4} - s_{1,3}s_{2,4}, \\
& s_{1,3}s_{1,4} - s_{2,3}s_{2,4} + s_{1,3}s_{3,4}, \\
& s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\
& s_{1,2}s_{1,4} - s_{2,2}s_{2,4} + s_{1,2}s_{3,4}, \\
& s_{2,3}^2 - s_{2,2}s_{3,3} - s_{2,4}^2 + s_{2,2}s_{4,4}, \\
& s_{1,3}s_{2,3} - s_{1,2}s_{3,3} - s_{1,4}s_{2,4} + s_{1,2}s_{4,4}, \\
& s_{1,3}^2 - s_{2,2}s_{3,3} + s_{1,3}s_{3,3} - s_{2,4}^2 + s_{2,2}s_{4,4}, \\
& s_{2,2}s_{1,3} - s_{1,2}s_{2,3}, \\
& s_{1,2}s_{1,3} - s_{2,2}s_{2,3} + s_{1,2}s_{3,3}, \\
& s_{1,2}^2s_{2,4} - s_{2,2}^2s_{2,4} + s_{1,2}s_{2,2}s_{3,4}, \\
& s_{1,2}^2s_{3,3} - s_{2,2}^2s_{3,3} + s_{1,2}s_{2,2}s_{3,3} - s_{1,2}s_{2,4}s_{3,4} - s_{1,2}^2s_{4,4} + s_{2,2}^2s_{4,4}, \\
& s_{1,2}^2s_{2,3} - s_{2,2}^2s_{2,3} + s_{1,2}s_{2,2}s_{3,3}.
\end{aligned}$$

One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
-0.106489268508094? & 3.7541451583602? & 1.186140661634507? & -0.0269749518305? & -1.686140661634507? & 1.18614066163451? & 0.042300079318905? & -3.423076923076923? & -1.686140661634507? \\
-1.089777942829606? & -0.27289947593211? & 0.?e-17 & -0.9556840077071? & 0.?e-18 & 0.?e-16 & -1.02508692203526? & 0.?e-17 & 0.?e-18 \\
1.089777942829606? & 0.49793045830773? & 0 & 0.9556840077071? & 0 & 2.76511155251136? & 1.02508692203526? & -12.61538461538462? & -14.82761155251136? \\
-0.089777942829606? & 1.7597769500383? & 1 & 0.0443159922929? & 1 & 1 & -0.025086922035259? & 1 & 1 \\
-0.918759450778702? & 0.072876564595505? & 0 & -0.11946050096339? & 0 & 0.?e-16 & 0.60794863996789? & 0.?e-17 & 0.?e-18 \\
0.918759450778702? & 0.62379469590986? & 0 & 0.1194605009634? & 0 & 2.33118351132236? & -0.60794863996789? & -1.576923076923077? & 8.79381648867764? \\
0.315199108720128? & 0.111640694699497? & 0 & 1.4335260115607? & 0 & 0 & 0.66453062100961? & 0 & 0 \\
-0.315199108720128? & -0.25652365595504? & 0 & -1.4335260115607? & 0 & -0.79975989842509? & -0.66453062100961? & 18.92307692307692? & 9.612259898425089?
\end{bmatrix}.$$

The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{11}x_{22}^2 + (-0.4654869009429914?)x_{13}^2x_{24}, \\
& x_{13}x_{14} + 3.241987216958221?x_{11}x_{22}, \\
& x_{14}^2 + (-(89787/18352))x_{11}x_{24}, \\
& x_{14}x_{22} + 1.509102582518676?x_{13}x_{24}, \\
& x_{14}x_{23} + 5.202493352150644?x_{11}x_{33}, \\
& x_{22}x_{23} + (-1.604723585872698?)x_{13}x_{33}, \\
& x_{23}x_{24} + 5.700800774557095?x_{11}x_{34}, \\
& x_{14}x_{33} + (-5.361105954350359?)x_{11}x_{34}, \\
& x_{22}x_{33} + 1.653648085441988?x_{13}x_{34}, \\
& x_{24}x_{33} + (-1.095782423671999?)x_{14}x_{34}, \\
& x_{33}^2 + (169/164)x_{23}x_{34}, \\
& x_{12}.
\end{aligned}$$

2. PA8 (Dimension of Lie Algebra: 21.)

The list of generators:

$$\begin{aligned}
& s_{1,1} - s_{2,2} + s_{1,3}, \\
& s_{3,3}s_{2,4} - s_{2,3}s_{3,4}, \\
& s_{1,4}^2 - s_{2,4}^2 + s_{1,4}s_{3,4}, \\
& s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\
& s_{2,3}s_{1,4} - s_{1,3}s_{2,4}, \\
& s_{1,3}s_{1,4} - s_{2,3}s_{2,4} + s_{1,3}s_{3,4}, \\
& s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\
& s_{1,2}s_{1,4} - s_{2,2}s_{2,4} + s_{1,2}s_{3,4}, \\
& s_{1,3}^2 - s_{2,3}^2 + s_{1,3}s_{3,3}, \\
& s_{2,2}s_{1,3} - s_{1,2}s_{2,3}, \\
& s_{1,2}s_{1,3} - s_{2,2}s_{2,3} + s_{1,2}s_{3,3}, \\
& s_{1,2}^2s_{2,4} - s_{2,2}^2s_{2,4} + s_{1,2}s_{2,2}s_{3,4}, \\
& s_{1,2}^2s_{2,3} - s_{2,2}^2s_{2,3} + s_{1,2}s_{2,2}s_{3,3}.
\end{aligned}$$

One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1.0000000000000? + 0.?e - 65 * I \\
0.?e - 13 & 1.110087446? & -0.507814351? & \frac{1}{8} & 0.?e - 11 & 0.?e - 9 & -1 & -0.41105832? & -0.2480911613979? + 0.?e - 65 * I \\
0.?e - 15 & 0.?e - 13 & -0.49218565? & \frac{-3}{16} & 0.?e - 10 & 0.?e - 10 & 0 & 0.?e - 11 & -0.751908838603? + 0.?e - 65 * I \\
0.?e - 14 & 0.?e - 13 & 0.384286943? & \frac{3}{32} & 0 & 0.?e - 11 & 0 & 0.?e - 11 & -0.963027100249? + 0.?e - 65 * I \\
1 & 1 & 0.507814351? & \frac{5}{16} & 1 & 1 & 1 & 1 & 0.2480911613979? + 0.?e - 65 * I \\
0.?e - 13 & 2.045146537? & 0.4921856496? & \frac{15}{16} & 0 & 0.?e - 9 & 0 & 1.78495055? & 0.7519088386021? + 0.?e - 65 * I \\
0 & -1.596802164? & -0.3842869428? & \frac{-21}{32} & 0 & 0 & 0 & 2.28612255? & 0.963027100249? + 0.?e - 65 * I \\
1.842329219213? & -4.090293074? & 0 & \frac{-21}{16} & 0 & \frac{-434233}{100000} & 0 & -3.5699011? & 0 \\
-1.438447187191? & 3.193604327? & 0 & \frac{15}{16} & 0 & \frac{-5561553}{1000000} & 0 & -4.5722451? & 0
\end{bmatrix}.$$

The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{12}^2 + 0.2167940331668742? x_{11}x_{13}, \\
& x_{12}x_{22} + 0.9008299337603309? x_{11}x_{24}, \\
& x_{13}x_{22} + (-4.155233982232942?) x_{12}x_{24}, \\
& x_{12}x_{23} + 0.3702936385364678? x_{11}x_{33}, \\
& x_{13}x_{23} + 0.648157776467543? x_{11}x_{34}, \\
& x_{23}x_{24} + (-0.4110583192886947?) x_{22}x_{33}, \\
& x_{12}x_{33} + 0.3794738117679905? x_{11}x_{34}, \\
& x_{13}x_{33} + (-1.750388634893360?) x_{12}x_{34}, \\
& x_{24}x_{33} + (-0.4212491143405442?) x_{22}x_{34}, \\
& x_{33}^2 + (-1.024791603949250?) x_{23}x_{34}, \\
& x_{14}.
\end{aligned}$$

3. PA11 (Dimension of Lie Algebra: 8.)

The list of generators:

$$\begin{aligned}
& s_{3,3}s_{2,4} - s_{2,3}s_{3,4}, \\
& s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\
& s_{2,3}s_{1,4} - s_{1,3}s_{2,4}, \\
& s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\
& s_{1,2}s_{1,4} - s_{1,1}s_{2,4} - s_{1,3}s_{2,4} + s_{1,2}s_{3,4}, \\
& s_{2,2}s_{1,3} - s_{1,2}s_{2,3}, \\
& s_{1,2}s_{1,3} - s_{1,1}s_{2,3} - s_{1,3}s_{2,3} + s_{1,2}s_{3,3}, \\
& s_{1,2}^2s_{2,4} - s_{1,1}s_{2,2}s_{2,4} - s_{1,2}s_{2,3}s_{2,4} + s_{1,2}s_{2,2}s_{3,4}, \\
& s_{1,2}^2s_{2,3} - s_{1,1}s_{2,2}s_{2,3} - s_{1,2}s_{2,2}^2 + s_{1,2}s_{2,2}s_{3,3}.
\end{aligned}$$

One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & -\frac{2}{5} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & \frac{2}{5} & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & \frac{1}{4} & 1 & 0 & 0 & 0 \\
1 & \frac{2}{5} & \frac{1}{4} & -1 & \frac{1}{10} & \frac{2}{5} & 0 & 0 & 0
\end{bmatrix}.$$

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The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{12}x_{13} - x_{11}x_{22}, \\
& x_{12}x_{14} + x_{11}x_{23}, \\
& x_{14}x_{22} + x_{13}x_{23}, \\
& x_{13}x_{24} + (-4)x_{11}x_{34}, \\
& x_{22}x_{24} + (-4)x_{12}x_{34}, \\
& x_{14}x_{33} + (-4)x_{13}x_{34}, \\
& x_{23}x_{33} + (4)x_{22}x_{34}.
\end{aligned}$$

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4. PB6 (Dimension of Lie Algebra: 53. The matrix is 10×10 .)

23

The list of generators:

$$\begin{aligned}
& s_{1,3} - s_{2,4}, \\
& s_{2,2} - s_{3,3}, \\
& s_{1,2} - s_{3,4}, \\
& s_{1,1} - s_{4,4}, \\
& s_{3,3}s_{2,4} - s_{2,3}s_{3,4}, \\
& s_{3,3}s_{1,4} - s_{2,4}s_{3,4}, \\
& s_{2,3}s_{1,4} - s_{2,4}^2.
\end{aligned}$$

24

5. PC8 (Dimension of Lie Algebra: 8.)

25

The list of generators:

$$\begin{aligned}
& s_{3,3}s_{2,4} - s_{2,3}s_{3,4}, \\
& s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\
& s_{2,3}s_{1,4} - s_{1,3}s_{2,4}, \\
& s_{1,3}^2 - s_{2,3}^2 - s_{1,1}s_{3,3} + s_{2,2}s_{3,3}, \\
& s_{1,3}s_{1,4} - s_{2,3}s_{2,4} - s_{1,1}s_{3,4} + s_{2,2}s_{3,4}, \\
& s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\
& s_{2,2}s_{1,3} - s_{1,2}s_{2,3}, \\
& s_{1,2}s_{1,3}s_{2,4} - s_{2,2}s_{2,3}s_{2,4} - s_{1,1}s_{2,2}s_{3,4} + s_{2,2}^2s_{3,4}, \\
& s_{1,2}s_{1,3}s_{2,3} - s_{2,2}s_{2,3}^2 - s_{1,1}s_{2,2}s_{3,3} + s_{2,2}^2s_{3,3}, \\
& s_{1,2}^2s_{2,3}s_{2,4} - s_{2,2}^2s_{2,3}s_{2,4} - s_{1,1}s_{2,2}^2s_{3,4} + s_{2,2}^3s_{3,4}, \\
& s_{1,2}^2s_{2,3}^2 - s_{2,2}^2s_{2,3}^2 - s_{1,1}s_{2,2}^2s_{3,3} + s_{2,2}^3s_{3,3}
\end{aligned}$$

26

One of the matrices indicating the change of variables is:

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 3 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & -3 & 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -3 & 1 & 0 & -3 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

27

The list of binomial generators after the change of variables is:

$$\begin{aligned} & x_{11}x_{14}^2x_{33} + (1/4)x_{12}^2x_{22}x_{34}, \\ & x_{11}x_{14}x_{24} + (-(1/4))x_{12}^2x_{34}, \\ & x_{13}^2x_{33} + (1/36)x_{11}x_{22}x_{34}, \\ & x_{13}x_{14}x_{33} + (1/12)x_{12}x_{22}x_{34}, \\ & x_{12}x_{13} + (-(1/3))x_{11}x_{14}, \\ & x_{12}x_{23} + (1/3)x_{11}x_{24}, \\ & x_{13}x_{23} + (1/36)x_{11}x_{34}, \\ & x_{14}x_{23} + (1/12)x_{12}x_{34}, \\ & x_{22}x_{23} - x_{13}x_{33}, \\ & x_{13}x_{24} + (-(1/12))x_{12}x_{34}, \\ & x_{22}x_{24} + x_{14}x_{33}. \end{aligned}$$

28

6. PC11 (The uncolored version: toric before the change of variables! Dimension of Lie Algebra: 13.)

29

30

The generators are:

$$\begin{aligned} & s_{3,3}s_{2,4} - s_{2,3}s_{3,4}, \\ & s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\ & s_{2,3}s_{1,4} - s_{1,3}s_{2,4}, \\ & s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\ & s_{2,2}s_{1,3} - s_{1,2}s_{2,3} \end{aligned}$$

31

7. PD1 = PD4 (All the edges are in the same color, while one of the ending vertex is in different colors. Dimension of Lie Algebra: 1.)

32

33

The list of generators:

$$\begin{aligned}
& s_{3,3} - s_{2,4} - s_{4,4}, \\
& s_{2,3} - s_{1,4} - s_{3,4}, \\
& s_{2,2} - s_{1,3} - s_{4,4}, \\
& s_{2,4}^2 - s_{1,4}s_{3,4} - s_{3,4}^2 + s_{2,4}s_{4,4}, \\
& s_{1,4}s_{2,4} - s_{1,3}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{1,3}s_{2,4} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,2}s_{2,4} - s_{1,1}s_{3,4} - s_{1,3}s_{3,4} + s_{1,2}s_{4,4}, \\
& s_{1,4}^2 - s_{1,2}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,3}s_{1,4} - s_{1,1}s_{3,4} - s_{1,3}s_{3,4} + s_{1,2}s_{4,4} + s_{1,4}s_{4,4}, \\
& s_{1,2}s_{1,4} - s_{1,1}s_{2,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,3}^2 - s_{1,1}s_{2,4} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} + 2s_{1,3}s_{4,4}, \\
& s_{1,2}s_{1,3} - s_{1,1}s_{1,4} - s_{1,1}s_{3,4} - s_{1,3}s_{3,4} + s_{1,2}s_{4,4} + s_{1,4}s_{4,4}, \\
& s_{1,2}^2 - s_{1,1}s_{1,3} - s_{1,2}s_{3,4} + s_{1,3}s_{4,4}.
\end{aligned}$$

34

One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\
0.6424183411? & 0.6785474106? & 0 & 6.90847655? & 0 & -0.19227483376? & 0 & 1.367263512? & -1.218088227? & 2.957689083427? \\
-0.5868775836? & -0.6031532539? & 0 & 0.801131023? & 0 & -0.26250441338? & 0 & -1.82401528? & 1.082745091? & -2.350443458008? \\
-1.0007607819? & -1.0313000738? & 0 & -4.44947121? & 0 & 0.27779739279? & 0 & 0.848255833? & -0.633240743? & 1.238889968061? \\
0.7971078239? & 0.7936934923? & 1 & 3.60226205? & 1 & 1.4749911733? & 1 & -40.122398390000001? & 4.16549019? & 1.195610889561? \\
-0.7265678439? & -0.7055053265? & -0.88888889? & 4.91801068? & \frac{9}{32} & 0.17104511807? & -0.8888888889? & -9.383533229999999? & -3.70265794? & -3.810034803257? \\
-1.2132238341? & -1.2063065077? & -1.51986443? & 1.60226205? & \frac{1}{2} & -0.5250088268? & 0.5198644306? & 18.707047450000001? & 2.165490181? & 1.592640485295? \\
0.1782626683? & 0.1905402385? & -0.51986443? & 4.40339307? & \frac{1}{2} & 1.2124867599? & 1.5198644306? & -20.546313229999999? & 5.24823528? & 0.642563904933? \\
0.3166663129? & 0.3257947474? & -0.888888889? & 9.367481890000001? & \frac{9}{32} & -0.10675227472? & -0.8888888889? & -11.4636247? & -3.069417196? & -2.77200602196? \\
1.4035038862? & 1.3968467461? & 1 & 2.80113103? & 1 & 1.7374955866? & 1 & -38.023952569999999? & 3.08274509? & -0.2934587092236?
\end{bmatrix}.$$

35

The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{12}x_{22}x_{24} + 10.913483364737968?x_{13}^2x_{34}, \\
& x_{12}x_{24}^2 + (-0.8632735864685306?)x_{13}x_{22}x_{34}, \\
& x_{13}x_{14} + (-0.1143755142204424?)x_{12}x_{22}, \\
& x_{14}^2 + 0.0434127868992438?x_{12}x_{23}, \\
& x_{14}x_{22} + 1.445932426688062?x_{12}x_{24}, \\
& x_{22}^2 + (1024/81)x_{13}x_{24}, \\
& x_{13}x_{23} + (-3.809459761574067?)x_{12}x_{24}, \\
& x_{14}x_{23} + (233/49)x_{12}x_{34}, \\
& x_{22}x_{23} + 41.57447573657725?x_{13}x_{34}, \\
& x_{23}^2 + (-109.53229176123576?)x_{14}x_{34}, \\
& x_{14}x_{24} + 1.248235271778149?x_{13}x_{34}, \\
& x_{23}x_{24} + (-3.288605990881599?)x_{22}x_{34}, \\
& x_{11}, \\
& x_{33}, \\
& x_{44}.
\end{aligned}$$

36 8. PD2 = PD3 (All the edges are in the same color, while one of the middle vertex is in
37 different colors. Dimension of Lie Algebra: 14.)

38 The list of generators:

$$\begin{aligned}
& s_{3,3} - s_{2,4} - s_{4,4}, \\
& s_{1,3} - s_{2,4}, \\
& s_{1,2} - s_{2,3} + s_{1,4}, \\
& s_{2,4}^2 - s_{2,3}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{1,4}s_{2,4} - s_{2,4}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{2,3}s_{2,4} - s_{2,2}s_{3,4} - s_{2,4}s_{3,4} + s_{2,3}s_{4,4}, \\
& s_{1,1}s_{2,4} - s_{2,3}s_{3,4} + s_{1,4}s_{3,4} + s_{3,4}^2 + s_{1,1}s_{4,4} - s_{2,4}s_{4,4} - s_{4,4}^2, \\
& s_{1,4}^2 - s_{3,4}^2 - s_{1,1}s_{4,4} + s_{2,4}s_{4,4} + s_{4,4}^2, \\
& s_{2,3}s_{1,4} - s_{2,3}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{2,2}s_{1,4} - s_{2,2}s_{3,4} + s_{2,3}s_{4,4} - s_{1,4}s_{4,4}, \\
& s_{2,3}^2 - s_{2,2}s_{2,4} - s_{2,3}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{1,1}s_{2,3} - s_{2,2}s_{3,4} + s_{2,4}s_{3,4} - s_{1,4}s_{4,4}, \\
& s_{1,1}s_{2,2} - s_{2,2}s_{2,4} + 2s_{2,3}s_{3,4} - s_{1,4}s_{3,4} - s_{3,4}^2 - s_{1,1}s_{4,4} - s_{2,2}s_{4,4} + s_{4,4}^2.
\end{aligned}$$

39 One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\
0.7e-10 & 0.7e-11 & \frac{1}{2} & 0.4524557? & 0 & 0.7e-13 & 0 & 0.3587351018134486? + 0.4979914981983616? * I & 0.3587351018134486? - 0.4979914981983616? * I & \frac{1}{2} \\
0.7e-10 & 0.7e-11 & 0.7e-11 & -0.5474276? & 0 & -0.692307692308? & 0 & 0.02662114706774843? - 0.775318715591643? * I & 0.02662114706774843? + 0.775318715591643? * I & 0 \\
0.7e-10 & 0.7e-11 & 0.7e-11 & -0.1282314? & 0 & 0.7e-13 & 0 & -0.3091915086478763? - 0.2775021994513020? * I & -0.3091915086478763? + 0.2775021994513020? * I & \frac{1}{2} \\
\frac{1}{4} & 0.7e-10 & 0.7e-11 & \frac{205867}{312500} & 0 & 0.692307692308? & 0 & 0.3390727005264831? + 0.2580267393136320? * I & 0.3390727005264831? - 0.2580267393136320? * I & 0 \\
0.7e-10 & 0.7e-11 & \frac{1}{2} & 0.4013535? & 0 & 0.7e-13 & 0 & -0.10756563964446617? - 1.346527093262331? * I & -0.10756563964446617? + 1.346527093262331? * I & 0 \\
0.7e-10 & 0.7e-11 & 0.7e-11 & -0.50320383? & 0 & -0.692307692308? & 0 & -1.353002273637889? + 0.03259240133389632? * I & -1.353002273637889? - 0.03259240133389632? * I & 0 \\
1 & 1 & 1 & 0.80614867? & 0 & 0.692307692308? & 0 & 0.1624533161633880? + 0.7706872472959516? * I & 0.1624533161633880? - 0.7706872472959516? * I & 0 \\
0.7e-11 & 0.7e-11 & 1.1111111111? & \frac{29123}{500000} & 1 & -0.276923076923? & 0 & -0.5637830761428779? - 0.3307316073402637? * I & -0.5637830761428779? + 0.3307316073402637? * I & \frac{5}{2} \\
1 & 1 & 1 & \frac{206421}{250000} & 0 & 1.384615384615? & 0 & 0.2847271847412152? + 0.2907825399171905? * I & 0.2847271847412152? - 0.2907825399171905? * I & 0
\end{bmatrix}$$

The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{11}x_{22}x_{24} + (-(10/9))x_{12}^2x_{44}, \\
& x_{11}x_{24}^2 + (10/9)x_{12}x_{22}x_{44}, \\
& x_{12}x_{13} + (-(9/10))x_{11}x_{22}, \\
& x_{13}^2 + (81/325)x_{11}x_{23}, \\
& x_{13}x_{22} + (9/10)x_{11}x_{24}, \\
& x_{22}^2 + x_{12}x_{24}, \\
& x_{12}x_{23} + (-(13/4))x_{11}x_{24}, \\
& x_{13}x_{23} + (-(13/4))x_{11}x_{44}, \\
& x_{22}x_{23} + (-(65/18))x_{12}x_{44}, \\
& x_{23}^2 + (4225/324)x_{13}x_{44}, \\
& x_{13}x_{24} - x_{12}x_{44}, \\
& x_{23}x_{24} + (65/18)x_{22}x_{44}, \\
& x_{14}, \\
& x_{33}, \\
& x_{34}.
\end{aligned}$$

9. PD5 (All the edges are in the same color, while the left half and the right half of the vertices share two more colors. Dimension of Lie Algebra: 25.)

The list of generators:

$$\begin{aligned}
& s_{3,3} - s_{2,4} - s_{4,4}, \\
& s_{1,1} - s_{2,2} + s_{1,3}, \\
& s_{2,4}^2 - s_{2,3}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{1,4}s_{2,4} - s_{1,3}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{2,3}s_{2,4} - s_{2,2}s_{3,4} - s_{2,4}s_{3,4} + s_{2,3}s_{4,4}, \\
& s_{1,3}s_{2,4} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,4}^2 - s_{2,3}s_{3,4} + s_{1,4}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{2,3}s_{1,4} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,3}s_{1,4} - s_{2,2}s_{3,4} + s_{1,3}s_{3,4} - s_{2,4}s_{3,4} + s_{2,3}s_{4,4}, \\
& s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\
& s_{1,2}s_{1,4} - s_{2,2}s_{2,4} + s_{1,2}s_{3,4}, \\
& s_{2,3}^2 - s_{2,2}s_{2,4} - s_{2,3}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{1,3}s_{2,3} - s_{1,2}s_{2,4} - s_{1,3}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{1,3}^2 - s_{2,2}s_{2,4} + s_{1,2}s_{3,4} - s_{2,3}s_{3,4} + s_{1,4}s_{3,4} + s_{2,4}s_{4,4}, \\
& s_{2,2}s_{1,3} - s_{1,2}s_{2,3}, \\
& s_{1,2}s_{1,3} - s_{2,2}s_{2,3} + s_{1,2}s_{2,4} + s_{1,2}s_{4,4}, \\
& s_{1,2}^2s_{3,4} - s_{2,2}^2s_{3,4} + s_{1,2}s_{2,3}s_{3,4} - s_{1,2}s_{3,4}^2 + s_{1,2}s_{2,4}s_{4,4} + s_{1,2}s_{4,4}^2, \\
& s_{1,2}^2s_{2,4} - s_{2,2}^2s_{2,4} + s_{1,2}s_{2,2}s_{3,4}, \\
& s_{1,2}^2s_{2,3} - s_{2,2}^2s_{2,3} + s_{1,2}s_{2,2}s_{2,4} + s_{1,2}s_{2,2}s_{4,4}.
\end{aligned}$$

One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
0 & 1.000000000000? + 0.?e - 63 * I & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\
1 & 0.?e - 12 + 0.?e - 63 * I & -0.43360955795? & 0.?e - 12 & 1 & 0 & 0 & 0 & 0 & 0.359761947003662? \\
0 & -1.000000000000? + 0.?e - 63 * I & -0.7645311821? & 0.?e - 12 & 0 & 0 & 0 & 0 & 0 & 0.55830083910934? \\
-1 & 0.?e - 13 + 0.?e - 63 * I & 0.1369383504? & 0.?e - 13 & 0 & 0 & 0 & 0 & 0 & -0.400433582425502? \\
0 & 0.?e - 13 + 0.?e - 64 * I & 0.18120334914? & 1 & 0 & 0 & 0 & 1 & 0 & -4.7342960554384? \\
0 & 0.?e - 11 + 0.?e - 62 * I & -8.311202236? & 0.?e - 11 & 0 & 1 & 0 & 0 & 0 & -0.262947832166867? \\
0 & 0.?e - 13 + 0.?e - 63 * I & -0.06048442541? & -1 & 0 & 0 & 0 & 0 & 0 & 2.37709368105542? \\
0 & 1.000000000000? + 0.?e - 63 * I & 0.6167916389? & 0 & 0 & 0 & 0 & 0 & 1 & 1.82522947842393? \\
1 & 0.?e - 13 + 0.?e - 64 * I & 0.003595009177? & 0.?e - 13 & 0 & 0 & 1 & 0 & 0.?e - 16 & 2.63881290818653? \\
0 & 1.000000000000? + 0.?e - 63 * I & 0.8616256762? & 1 & 0 & 0 & 0 & 0 & 1 & -1.4780140242273?
\end{bmatrix}.$$

The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{11}x_{33}^2 - x_{12}x_{22}x_{34}, \\
& x_{12}x_{33}^2 + x_{22}^2x_{34}, \\
& x_{12}^2 + x_{11}x_{22}, \\
& x_{12}x_{14} + x_{11}x_{23}, \\
& x_{14}^2 + x_{11}x_{24}, \\
& x_{14}x_{22} - x_{11}x_{33}, \\
& x_{12}x_{23} - x_{11}x_{33}, \\
& x_{14}x_{23} - x_{11}x_{34}, \\
& x_{22}x_{23} + x_{12}x_{33}, \\
& x_{23}^2 + x_{12}x_{34}, \\
& x_{12}x_{24} - x_{11}x_{34}, \\
& x_{22}x_{24} + x_{12}x_{34}, \\
& x_{23}x_{24} + x_{14}x_{34}, \\
& x_{14}x_{33} - x_{12}x_{34}, \\
& x_{23}x_{33} - x_{22}x_{34}, \\
& x_{24}x_{33} - x_{23}x_{34}, \\
& x_{13}, \\
& x_{44}.
\end{aligned}$$

10. PD7 (All the edges are in the same color, while the odd half and the even half of the

vertices share two more colors. Dimension of Lie Algebra: 43.)

The list of generators:

$$\begin{aligned}
& s_{2,3} - s_{1,4} - s_{3,4}, \\
& s_{2,2} - s_{2,4} - s_{4,4}, \\
& s_{1,2} - s_{3,4}, \\
& s_{1,1} + s_{1,3} - s_{3,3}, \\
& s_{1,4}s_{2,4} - s_{2,4}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{3,3}s_{2,4} - s_{1,4}s_{3,4} - s_{3,4}^2, \\
& s_{1,3}s_{2,4} - s_{1,4}s_{3,4} - s_{3,4}^2 + s_{1,3}s_{4,4}, \\
& s_{1,4}^2 - s_{3,4}^2 + s_{1,3}s_{4,4}, \\
& s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\
& s_{1,3}s_{1,4}s_{3,4} - s_{3,3}s_{3,4}^2 + s_{1,3}s_{3,3}s_{4,4}, \\
& s_{1,3}^2s_{3,4}^2 - s_{3,3}^2s_{3,4}^2 + s_{1,3}s_{3,3}^2s_{4,4}
\end{aligned}$$

One of the matrices indicating the change of variables is:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1.450606195? & 3.657794008336043? + 2.998113178616908? * I & 3.657794008336043? - 2.998113178616908? * I & \frac{29331923}{250000} & 0.?e-15 & 0.?e-16 & 1 & 1 & 0 & 0 \\ 2.13713366998? & -0.07539690902259121? + 1.809762932781717? * I & -0.07539690902259121? - 1.809762932781717? * I & 1.49921129? & -\frac{1}{2} & 0.?e-15 & 0.?e-15 & 0.?e-15 & 0 & 0 \\ -1.15355442849? & -2.009727985133510? - 0.6590970572023312? * I & -2.009727985133510? + 0.6590970572023312? * I & 9.2000122999999999? & 0.?e-15 & 0.?e-15 & -1 & 0.?e-14 & 0 & 0 \\ 0.82756377579? & 2.176211070270429? + 0.837431704864615? * I & 2.176211070270429? - 0.837431704864615? * I & -\frac{34879273}{1000000} & 0.?e-15 & 0.?e-15 & 0.?e-15 & 0.?e-15 & 1 & 1 \\ -1.11569678295? & -0.985343842738591? - 6.087273484887979? * I & -0.985343842738591? + 6.087273484887979? * I & \frac{149009}{1000000} & 0.?e-15 & 0.?e-15 & 0.?e-14 & 1 & 0.?e-14 & 0.?e-13 \\ -4.9852469607? & 3.160574880580981? - 0.4593293207956499? * I & 3.160574880580981? + 0.4593293207956499? * I & \frac{6552257}{1000000} & 0 & 0 & 0.?e-15 & 0.?e-15 & -1 & 0.?e-13 \\ -3.0694843805? & -0.6160541523654867? - 1.929143561912773? * I & -0.6160541523654867? + 1.929143561912773? * I & -8.052379500000001? & \frac{1}{2} & 1 & 0 & 0 & 0.?e-15 & 0.?e-15 \\ 0.67938815951? & -0.900654077089071? - 1.851974961355472? * I & -0.900654077089071? + 1.851974961355472? * I & \frac{118310841}{1000000} & 0 & 0 & 1 & 1 & 0 & 0 \\ 3.773246224? & 2.487127564938216? - 4.081630246686257? * I & 2.487127564938216? + 4.081630246686257? * I & -\frac{141308823}{1000000} & 0 & 0 & 0 & 0 & 2 & 1 \end{bmatrix}$$

The list of binomial generators after the change of variables is:

$$\begin{aligned} & x_{22}x_{34}^2 + (2)x_{24}^2x_{44}, \\ & x_{23}x_{24} + (-(1/2))x_{22}x_{33}, \\ & x_{24}x_{33} + (1/2)x_{22}x_{34}, \\ & x_{33}^2 + (1/2)x_{22}x_{44}, \\ & x_{23}x_{34} + (-(1/2))x_{22}x_{44}, \\ & x_{33}x_{34} - x_{24}x_{44}, \\ & x_{11}, \\ & x_{12}, \\ & x_{13}, \\ & x_{14}. \end{aligned}$$

(The matrix returned immediately, but it took about 4 minutes for the Sage algorithms to return the set of polynomials.)

11. PD10 (All the edges are in the same color, while all vertices in different colors. Dimension of Lie Algebra: 14. The matrix is 10×10 .)

$$\begin{aligned}
& s_{2,3} - s_{1,4} - s_{3,4}, \\
& s_{1,4}s_{2,4} - s_{2,2}s_{3,4} + s_{1,4}s_{4,4} + s_{3,4}s_{4,4}, \\
& s_{3,3}s_{2,4} - s_{1,4}s_{3,4} - s_{3,4}^2, \\
& s_{1,3}s_{2,4} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,4}^2 - s_{1,2}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{3,3}s_{1,4} - s_{1,3}s_{3,4}, \\
& s_{2,2}s_{1,4} - s_{1,2}s_{2,4}, \\
& s_{1,2}s_{1,4} - s_{1,1}s_{2,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{2,2}s_{3,3} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} - s_{3,3}s_{4,4}, \\
& s_{1,2}s_{3,3} - s_{1,1}s_{3,4} - s_{1,3}s_{3,4}, \\
& s_{2,2}s_{1,3} - s_{1,1}s_{2,4} - s_{1,2}s_{3,4} - s_{1,4}s_{3,4} + s_{1,3}s_{4,4}, \\
& s_{1,2}s_{1,3} - s_{1,1}s_{1,4} - s_{1,3}s_{1,4}, \\
& s_{1,2}^2 - s_{1,1}s_{2,2} - s_{1,2}s_{3,4} + s_{1,1}s_{4,4} + s_{1,3}s_{4,4}, \\
& s_{1,3}s_{1,4}s_{3,4} - s_{1,1}s_{3,4}^2 - s_{1,3}s_{3,4}^2 + s_{1,3}s_{3,3}s_{4,4}, \\
& s_{1,2}s_{2,4}^2 - s_{2,2}s_{3,4}^2 + s_{1,2}s_{2,4}s_{4,4} + s_{2,2}s_{3,4}s_{4,4}, \\
& s_{1,1}s_{2,4}^2 - s_{1,2}s_{2,2}s_{3,4} + s_{2,2}s_{3,4}^2 + s_{1,1}s_{2,4}s_{4,4} - s_{1,4}s_{3,4}s_{4,4} - s_{3,4}^2s_{4,4}, \\
& s_{1,3}s_{3,4}^2 - s_{1,1}s_{3,3}s_{3,4}^2 - s_{1,3}s_{3,3}s_{3,4}^2 + s_{1,3}s_{3,3}^2s_{4,4}
\end{aligned}$$

55

One of the matrices indicating the change of variables is:

$$\begin{bmatrix}
1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{96}{5} \\
0 & \frac{-1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 6 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -14 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -7 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 21 \\
0 & \frac{1}{2} & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -3 \\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 8 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & -14
\end{bmatrix}$$

The list of binomial generators after the change of variables is:

$$\begin{aligned}
& x_{11}x_{14}^2 + (1/4)x_{12}^2x_{22}, \\
& x_{12}x_{13} + (-2)x_{11}x_{14}, \\
& x_{13}^2 + x_{11}x_{22}, \\
& x_{13}x_{14} + (1/2)x_{12}x_{22}, \\
& x_{13}x_{23} - x_{11}x_{24}, \\
& x_{14}x_{23} + (-(1/2))x_{12}x_{24}, \\
& x_{22}x_{23} - x_{11}x_{34}, \\
& x_{13}x_{24} + x_{11}x_{34}, \\
& x_{14}x_{24} + (1/2)x_{12}x_{34}, \\
& x_{22}x_{24} - x_{13}x_{34}, \\
& x_{24}^2 + x_{23}x_{34}, \\
& x_{12}x_{33} + (-2)x_{11}x_{34}, \\
& x_{14}x_{33} - x_{13}x_{34}, \\
& x_{44}.
\end{aligned}$$

12. PD12 (All vertices in one color, all edges in another color. Determining if it's toric using the Sage codes has taken more than 80 hours, but no results were returned yet.)

However, we are confident that this case is binomial. First, the Macaulay codes showed that the ideal I is generated by the 9 polynomials with 10 variables below:

$$\begin{aligned}
& s_{3,3} - s_{2,4} - s_{4,4}, \\
& s_{2,3} - s_{1,4} - s_{3,4}, \\
& s_{1,3} - s_{2,4}, \\
& s_{2,2} - s_{2,4} - s_{4,4}, \\
& s_{1,2} - s_{3,4}, \\
& s_{1,1} - s_{4,4}, \\
& s_{2,4}^2 - s_{1,4}s_{3,4} - s_{3,4}^2 + s_{2,4}s_{4,4}, \\
& s_{1,4}s_{2,4} - s_{2,4}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{1,4}^2 - s_{3,4}^2 + s_{2,4}s_{4,4}
\end{aligned}$$

Although the Sage codes did not tell us whether the ideal can be binomial after some change of variables, once we eliminate the binomial generators, namely getting a new

63 ideal I' with 7 variables, generated by:

$$\begin{aligned}
& s_{3,3} - s_{2,4} - s_{4,4}, \\
& s_{2,3} - s_{1,4} - s_{3,4}, \\
& s_{2,2} - s_{2,4} - s_{4,4}, \\
& s_{2,4}^2 - s_{1,4}s_{3,4} - s_{3,4}^2 + s_{2,4}s_{4,4}, \\
& s_{1,4}s_{2,4} - s_{2,4}s_{3,4} + s_{1,4}s_{4,4}, \\
& s_{1,4}^2 - s_{3,4}^2 + s_{2,4}s_{4,4}.
\end{aligned}$$

64 The Sage code told us that new ideal is binomial after the change of variables. Below
65 is the (reduced) Groebner basis with 7 new variables:

$$(x_{22}^2 - 3x_{14}x_{23}, x_{22}x_{23} - 9x_{14}x_{34}, x_{23}^2 - 3x_{22}x_{34}, x_{24}, x_{33}, x_{44})$$

66 This 7×7 matrix, calculated by the Sage codes indicates one way to change the variables
67 and make the new ideal binomial:

$$M' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -49.012234? & -0.986258? & -6 & \frac{50}{21} & \frac{46}{147} & -0.011676? & -\frac{90}{343} \\ 33.243205? & -0.411454? & 4 & \frac{20}{21} & -\frac{52}{147} & 0.156996? & \frac{4}{49} \\ -12.863897? & 1.508237? & -2 & -\frac{26}{21} & -\frac{10}{21} & -0.192562? & \frac{2}{7} \\ -43.337092? & -0.776085? & -6 & \frac{50}{21} & \frac{46}{147} & -0.685470? & -\frac{90}{343} \\ 24.106074? & 4.162263? & 3 & -\frac{1}{21} & -\frac{199}{147} & -1.206264? & -\frac{45}{49} \\ -25.777955? & 0.624403? & -4 & \frac{76}{21} & \frac{116}{147} & 0.124980? & -\frac{188}{343} \end{bmatrix}.$$

68 Max offered a smart way to extend it to a 10×10 matrix:

$$M = \begin{bmatrix} 1 & 0 & 0 & -25.777955? & 0.624403? & -4 & \frac{76}{21} & \frac{116}{147} & 0.124980? & -\frac{188}{343} \\ 0 & 1 & 0 & 24.106074? & 4.162263? & 3 & -\frac{1}{21} & -\frac{199}{147} & -1.206264? & -\frac{45}{49} \\ 0 & 0 & 1 & -12.863897? & 1.508237? & -2 & -\frac{26}{21} & -\frac{10}{21} & -0.192562? & \frac{2}{7} \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & -49.012234? & -0.986258? & -6 & \frac{50}{21} & \frac{46}{147} & -0.011676? & -\frac{90}{343} \\ 0 & 0 & 0 & 33.243205? & -0.411454? & 4 & \frac{20}{21} & -\frac{52}{147} & 0.156996? & \frac{4}{49} \\ 0 & 0 & 0 & -12.863897? & 1.508237? & -2 & -\frac{26}{21} & -\frac{10}{21} & -0.192562? & \frac{2}{7} \\ 0 & 0 & 0 & -43.337092? & -0.776085? & -6 & \frac{50}{21} & \frac{46}{147} & -0.685470? & -\frac{90}{343} \\ 0 & 0 & 0 & 24.106074? & 4.162263? & 3 & -\frac{1}{21} & -\frac{199}{147} & -1.206264? & -\frac{45}{49} \\ 0 & 0 & 0 & -25.777955? & 0.624403? & -4 & \frac{76}{21} & \frac{116}{147} & 0.124980? & -\frac{188}{343} \end{bmatrix}.$$

69 Now, we use $\vec{S}$ to generate the new variables x_{11}, x_{12}, x_{13} . Let $\vec{s} = [s_{1,1}, \dots, s_{4,4}]^T$,
70 and $\vec{x} = [x_{11}, \dots, x_{44}]^T$. Naming the vectors with 7 entries $\vec{s}' = [s_{1,4}, \dots, s_{4,4}]^T$, and
71 $\vec{x}' = [x_{14}, \dots, x_{44}]^T$, we know that $\vec{s}' = M'\vec{x}'$ demonstrated a way to make I' binomial
72 with the new set of variables.

The way we wrote M ensures that I' can also be binomial with new variables $\tilde{x}$, and that we can get the variables similarly from $\tilde{s} = M\tilde{x}$. It works because we map the binomials we deleted earlier, namely $s_{1,1} - s_{4,4}$, $s_{1,2} - s_{3,4}$, $s_{1,3} - s_{2,4}$ into monomials x_{11}, x_{12}, x_{13} with the matrix multiplication, so the original ideal has to also be binomial.

References

- [1] Jane I Coons, Aida Maraj, Pratik Misra, and Miruna-Ştefana Sorea. Symmetrically colored gaussian graphical models with toric vanishing ideals. *SIAM Journal on Applied Algebra and Geometry*, 7(1):133–158, 2023.